

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

December 2020**End-Semester Examination****Max. Marks:** 30**Duration:** 1hr.15 min (attempting questions) +15 min (uploading)**Class:** SY B. Tech**Name of the course:** Integral Transform and Vector Calculus**Branch:** Computer Engineering**Semester :** III**Course Code:** 2UCC301

Q.no.	Sub Q.no.	Question	10Marks
1	A	Evaluate $\int_0^\infty \frac{\cos(3t) - \cos(2t)}{t} dt$	5
		OR	
		Solve $\frac{d^2 y}{dt^2} + 4y = H(t-2)$ with conditions $y(0) = 0, y'(0) = 1$	5
	B	Find Z-Transform of $\{3^k \cosh ak\}$, $k \geq 0$ and also find its ROC.	5
		OR	
		Find Inverse Z-Transform of $\frac{1}{(z-3)(z-2)}$, $ z < 2$	5
2		Find Fourier series to represent $f(x) = 2x - x^2$ in $(0,3)$ Hence prove that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$	10
3	A	Find the constants a and b so that the surface $ax^2 - byz = (a+2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at $(1,-1,2)$	5
	B	Find the scalar potential for $\vec{F} = (2xyz^2)\hat{i} + (x^2z^2 + z \cos yz)\hat{j} + (2x^2yz + y \cos yz)\hat{k}$ if it exists and find work done in moving an object in this field from $(0, 0, 1)$ to $(1, \pi/4, 2)$.	5
		OR	
		State Stoke's Theorem and evaluate $\int (xydx + xy^2dy)$ over the square in xy-plane with vertices $(1,0), (0,1), (-1,0)$ and $(0,-1)$	5